GlobalProtect Connect Before Logon (PLAP) — Test-Window Diagnostic Runbook

Goal: on a clean **Windows 11** Autopilot VM, reproduce the failure, confirm whether the Connect-Before-Logon tile appears at sign-in, and **capture IME + Autopilot + GlobalProtect logs before the VM is wiped.** Read-only/collection tools only. Vendor portal/gateway & app internals per Palo Alto docs.

0 · Before the window

• Fresh **Windows 11** VM on the **Hybrid AAD Join** Autopilot profile (not Win10).

• GlobalProtect MSI ≥ **5.2**, portal FQDN, auth method. The five GPPLAP-*.ps1 scripts on the VM.

• App `GlobalProtect – Connect Before Logon (PLAP) – Win32` assigned to `PILOT – GlobalProtect PLAP – Autopilot Devices` and set as an **ESP blocking app**.

1 · Provision & watch the ESP app phase

Start user-driven provisioning. If it hangs/red-X's on the GlobalProtect app, **do not reset** — go to step 2.

2 · Capture everything (evidence first — the critical step)

```
powershell -ExecutionPolicy Bypass -File .\GPPLAP-Collect-Logs.ps1 -OutFolder C:\Temp\gpplap
```

Run **elevated**. Produces one timestamped ZIP: IME logs (IntuneManagementExtension / AgentExecutor / AppWorkload), the MdmDiagnosticsTool Autopilot ZIP, PanGPS/PanGPA logs, exported Event channels, and summary.txt (exit codes + PanSetup state). **Copy the ZIP off the VM now.**

3 · Confirm PLAP state

```
powershell -ExecutionPolicy Bypass -File .\GPPLAP-Validation.ps1
```

PASS requires **GP ≥ 5.2 AND Prelogon = 1 AND PLAP registered**. Also prints dsregcmd join type (DomainJoined + AzureAdJoined = Hybrid).

4 · Confirm the sign-in tile (ground truth)

Lock / sign out → click the **network sign-in / Connect Before Logon** tile. If **absent**, PLAP is not registered → decision tree.

Decision tree — read summary.txt first

What you see	Most likely cause	Fix
"Unzipping" / exit 0x87D30067 in IME log	IME can't extract .intunewin into IMECache — AV/Defender lock, low disk, or bad/nested re-wrap	Exclude IME Content/IMECache from AV during provisioning; free disk; re-wrap cleanly (don't nest an already-wrapped payload)
Install loops / IME state lost after reboot	Forced Restart-Computer inside the install script	Remove scripted reboot; return 3010 (ESP owns reboot); Start-Process -Wait; codes 0/1707=ok, 3010=reboot, 1618=retry
No CBL tile, app 'installed'	GP < 5.2, Prelogon not set, or -registerplap didn't run in SYSTEM / ran before PanGPS ready	GP ≥ 5.2; set PanSetup\Prelogon=1 + Portal; run -registerplap as SYSTEM after a service-readiness wait
Worked on Win10, fails now	Target OS mismatch	Re-test on Windows 11 (production target)
Tile shows, tunnel up, hybrid first-logon fails	DC not reachable via gateway / ODJ / time skew	Confirm gateway→DC line-of-sight, Offline Domain Join blob, VM clock
ESP never waits for the app	App not marked ESP blocking	Add app to ESP blocking-apps list
Enrollment-time device group empty	Assigned group missing Provisioning Client SP owner	Set SP AppId f1346770-5b25-470b-88bd-d5744ab7952c as an owner of the Assigned group

Enable reference (the fix, per Palo Alto docs)

• Install GlobalProtect (≥ 5.2).

• Registry — HKLM\SOFTWARE\Palo Alto Networks\GlobalProtect\PanSetup\ → Prelogon=1, set Portal.

• Wait for the PanGPS service, then as SYSTEM: PanGPS.exe -registerplap (from C:\Program Files\Palo Alto Networks\GlobalProtect).

• Package all of the above as ONE Win32 app that returns 3010 for reboot (no scripted restart).

Exit criterion: log ZIP captured off-box, GPPLAP-Validation.ps1 shows the exact failing precondition, and you know which decision-tree row you're in — enough to stabilise the next build without another full test window.